

MEMO

WTW158 Calculus

Class Test 3 [Mo]

15 April 2024

Time: 15 min

8

Surname:

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Initials:

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Student number:

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Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks] Evaluate $\lim_{x \rightarrow 0} \frac{3x}{\sqrt{5} - \sqrt{x+5}}$. [Do not use L'Hospital's rule]

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{3x}{\sqrt{5} - \sqrt{x+5}} \quad \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{3x (\sqrt{5} + \sqrt{x+5})}{(\sqrt{5} - \sqrt{x+5})(\sqrt{5} + \sqrt{x+5})} \\ &= \lim_{x \rightarrow 0} \frac{3x (\sqrt{5} + \sqrt{x+5})}{5 - (x+5)} \\ &= \lim_{x \rightarrow 0} \frac{3x (\sqrt{5} + \sqrt{x+5})}{-x} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} (-3)(\sqrt{5} + \sqrt{x+5}) \\ &= -3(\sqrt{5} + \sqrt{5}) \\ &= -6\sqrt{5} \end{aligned}$$

QUESTION 2 [2 marks]

Find the following limit: $\lim_{x \rightarrow 0} \frac{\tan^3(2x)}{x^3}$. [Do not use L'Hospital's rule]

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\tan^3(2x)}{x^3} \quad \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{\tan 2x}{x} \cdot \frac{\tan 2x}{x} \cdot \frac{\tan 2x}{x} \\ &= \lim_{x \rightarrow 0} \frac{\tan 2x}{2x} \cdot \frac{\tan 2x}{2x} \cdot \frac{\tan 2x}{2x} \times 8 \\ &= 8 \left(\lim_{x \rightarrow 0} \frac{\tan 2x}{2x} \right) \left(\lim_{x \rightarrow 0} \frac{\tan 2x}{2x} \right) \left(\lim_{x \rightarrow 0} \frac{\tan 2x}{2x} \right) \\ &= 8 \times 1 \times 1 \times 1 \\ &= 8 \end{aligned}$$

NB!!

must split
the limits!

OR:

$\tan^3 2x$
can be written
as $\frac{\sin^3 2x}{\cos^3 2x}$

↑
please
check
steps

QUESTION 3 [1 marks]

Define the continuity of a function $f(x)$ at a point $x = a$.

Definition of continuity.

f is continuous at $x=a$ if $\lim_{x \rightarrow a} f(x) = f(a)$ (1)

QUESTION 4 [3 marks] Determine a and b so that the function

$$f(x) = \begin{cases} x^2, & \text{if } -2 \leq x \leq 3 \\ ax + b, & \text{if } x < -2 \text{ or } x > 3, \end{cases}$$

is continuous at $x = -2$ and $x = 3$.

At $x = -2$: $f(-2) = 4$

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (ax + b) = -2a + b$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} x^2 = (-2)^2 = 4 = f(-2)$$

f continuous at $x = -2$
 $\Rightarrow -2a + b = 4$ (1)

At $x = 3$: $f(3) = 9$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} x^2 = 3^2 = 9 = f(3)$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} ax + b = 3a + b$$

f cont at $x = 3 \Rightarrow 3a + b = 9$ (2)

Solve (1) and (2)

$$\begin{aligned} -2a + b &= 4 \\ 3a + b &= 9 \end{aligned}$$

$$5a = 5 \Rightarrow a = 1, b = 6$$

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Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks]

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{5} - \sqrt{x+5}}{5x}$. Do not use L'Hospital's rule.

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{\sqrt{5} - \sqrt{x+5}}{5x} \quad \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{\sqrt{5} - \sqrt{x+5}}{5x} \left(\frac{\sqrt{5} + \sqrt{x+5}}{\sqrt{5} + \sqrt{x+5}} \right) \\
 &= \lim_{x \rightarrow 0} \frac{5 - (x+5)}{5x(\sqrt{5} + \sqrt{x+5})} = \lim_{x \rightarrow 0} \frac{-x}{5x(\sqrt{5} + \sqrt{x+5})} \\
 &= \lim_{x \rightarrow 0} \frac{-1}{5(\sqrt{5} + \sqrt{x+5})} = \frac{-1}{10\sqrt{5}}
 \end{aligned}$$

QUESTION 2 [2 marks]

Find the following limit: $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(4x)}$. Do not use L'Hospital's rule.

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 4x} \quad \left(\frac{0}{0} \right) \\
 &= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{4x}{\sin 4x} \times \frac{3}{4} \\
 &= \frac{3}{4} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \lim_{x \rightarrow 0} \frac{4x}{\sin 4x} \\
 &= \frac{3}{4} \times 1 \times 1 \\
 &= \frac{3}{4}
 \end{aligned}$$

QUESTION 3 [1.5 marks]

Name 3 things a function $f(x)$ requires to be continuous at a point $x = a$.

- ① $f(a)$ must exist
- ② $\lim_{x \rightarrow a} f(x)$ must exist
- ③ $\lim_{x \rightarrow a} f(x) = f(a)$.

QUESTION 4 [2.5 marks]

Determine c and b so that the function

$$f(x) = \begin{cases} x^2, & \text{if } -1 \leq x \leq 2 \\ cx + b, & \text{if } x < -1 \text{ or } x > 2, \end{cases}$$

is continuous at $x = -1$ and $x = 2$.

$$f(-1) = (-1)^2 = 1$$
$$f(2) = (2)^2 = 4.$$

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} cx + b = -c + b$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} x^2 = (-1)^2 = 1$$

$$f \text{ continuous at } x = -1 \Rightarrow \boxed{-c + b = 1} = f(-1)$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} x^2 = 4$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} cx + b = 2c + b$$

$$f \text{ cont. at } x = 2 \Rightarrow \boxed{2c + b = 4} = f(2)$$

Solve simultaneously:

$$\begin{aligned} 1 &= -c + b & \text{①} \\ 4 &= 2c + b & \text{②} \end{aligned}$$

$$\text{②} - \text{①} \quad 3 = 3c$$

$$\Rightarrow c = 1 \text{ and } b = 2.$$

for f to be continuous, we need $b = 2, c = 1$.

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Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks]

Evaluate $\lim_{x \rightarrow 1} \arcsin\left(\frac{1-\sqrt{x}}{1-x}\right)$.

$$\begin{aligned} & \lim_{x \rightarrow 1} \arcsin\left(\frac{1-\sqrt{x}}{1-x}\right) \\ &= \arcsin \lim_{x \rightarrow 1} \left(\frac{1-\sqrt{x}}{1-x}\right) \\ &= \arcsin \lim_{x \rightarrow 1} \frac{(1-\sqrt{x})}{(1-\sqrt{x})(1+\sqrt{x})} = \arcsin \lim_{x \rightarrow 1} \frac{1}{1+\sqrt{x}} \\ &= \arcsin \frac{1}{2} \\ &= \frac{\pi}{6}. \end{aligned}$$

QUESTION 2 [2 marks]

Evaluate $\lim_{x \rightarrow 0} \frac{\tan 5x}{5x - \sin x}$. Do not use the L'Hospital's rule.

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\tan 5x}{5x - \sin x} \quad \left(\frac{0}{0}\right) \\ &= \lim_{x \rightarrow 0} \frac{\tan 5x}{5x} \cdot \frac{5x}{5x - \sin x} \\ &= \lim_{x \rightarrow 0} \frac{\tan 5x}{5x} \cdot \frac{1}{1 - \left(\frac{\sin x}{x}\right)\frac{1}{5}} \\ &= \frac{\lim_{x \rightarrow 0} \frac{\tan 5x}{5x}}{1 - \frac{1}{5} \lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1 - \frac{1}{5}} = \frac{1}{\frac{4}{5}} = \frac{5}{4}. \end{aligned}$$

QUESTION 3 [2 marks]

If $\sqrt{7-2x^2} \leq f(x) \leq \sqrt{7-x^2}$ for $-1 \leq x \leq 1$, find $\lim_{x \rightarrow 0} f(x)$.

$$\lim_{x \rightarrow 0} \sqrt{7-2x^2} = \sqrt{7}$$

$$\lim_{x \rightarrow 0} \sqrt{7-x^2} = \sqrt{7}$$

$\Rightarrow$ By the squeeze theorem,

$$\lim_{x \rightarrow 0} f(x) = \sqrt{7}.$$

QUESTION 4 [2 marks]

Determine the value of k so that the function

$$g(x) = \begin{cases} x - k, & \text{if } x < 3 \\ 1 - kx, & \text{if } x \geq 3, \end{cases}$$

is continuous at $x = 3$.

f continuous at $x = 3$

$$\Rightarrow \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3) = 1 - 3k.$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} x - k = 3 - k.$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 1 - kx = 1 - 3k.$$

$$\begin{aligned} \Rightarrow 3 - k &= 1 - 3k \\ \Rightarrow 2 &= -2k \\ \Rightarrow \underline{\underline{k}} &= \underline{\underline{-1}} \end{aligned} \quad \left. \vphantom{\begin{aligned} \Rightarrow 3 - k &= 1 - 3k \\ \Rightarrow 2 &= -2k \\ \Rightarrow \underline{\underline{k}} &= \underline{\underline{-1}} \end{aligned}} \right\} \Rightarrow f \text{ is cont. at } x = 3 \text{ if } \underline{\underline{k = -1}}.$$

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QUESTION 1 [2 marks]

Determine the interval on which the function $f(x) = \frac{\ln x + \arcsin x}{(x-1)}$ is continuous. Show all the steps.

$\ln x$ is continuous on $(0, \infty)$

$\arcsin x$ is continuous on $(-1, 1]$

$\Rightarrow \ln x + \arcsin x$ is continuous on $(0, 1]$.

$x-1$ is a polynomial, so continuous on $\mathbb{R}$

$\frac{\ln x + \arcsin x}{x-1}$ is continuous on $(0, 1)$, since $x-1 \neq 0 \Rightarrow x \neq 1$.

QUESTION 2 [2 marks]

Evaluate $\lim_{h \rightarrow 0} \frac{h}{\sqrt{x+5h} - \sqrt{x}}$. (Do not use the L'Hospital's rule.)

$$\lim_{h \rightarrow 0} \frac{h}{\sqrt{x+5h} - \sqrt{x}} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{h \rightarrow 0} \frac{h}{\sqrt{x+5h} - \sqrt{x}} \cdot \left(\frac{\sqrt{x+5h} + \sqrt{x}}{\sqrt{x+5h} + \sqrt{x}} \right)$$

$$= \lim_{h \rightarrow 0} \frac{h(\sqrt{x+5h} + \sqrt{x})}{x+5h - x} = \lim_{h \rightarrow 0} \frac{h(\sqrt{x+5h} + \sqrt{x})}{5h}$$

$$= \frac{1}{5} \lim_{h \rightarrow 0} (\sqrt{x+5h} + \sqrt{x}) = \frac{2\sqrt{x}}{5}$$

QUESTION 3 [2 marks]

If $3 - x^2 \leq g(x) \leq 3 \cos x$ for all x , find $\lim_{x \rightarrow 0} g(x)$.

$$\lim_{x \rightarrow 0} (3 - x^2) = 3$$

$$\lim_{x \rightarrow 0} 3 \cos x = 3$$

$\Rightarrow$ By the squeeze theorem

$$\lim_{x \rightarrow 0} g(x) = 3.$$

QUESTION 4 [2 marks]

Determine the value of k so that the function

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 2 \\ k - x^2, & \text{if } x > 2, \end{cases}$$

is continuous at $x = 2$.

f cont. at $x = 2$

$$\Rightarrow \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2).$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} x^2 = 4$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} k - x^2 = k - 4$$

$$f(2) = 4$$

$$\Rightarrow 4 = k - 4 = f(2)$$

$$\Rightarrow \underline{\underline{k = 8}}$$

~~f is conti~~ If $k = 8$, f is continuous at $x = 2$